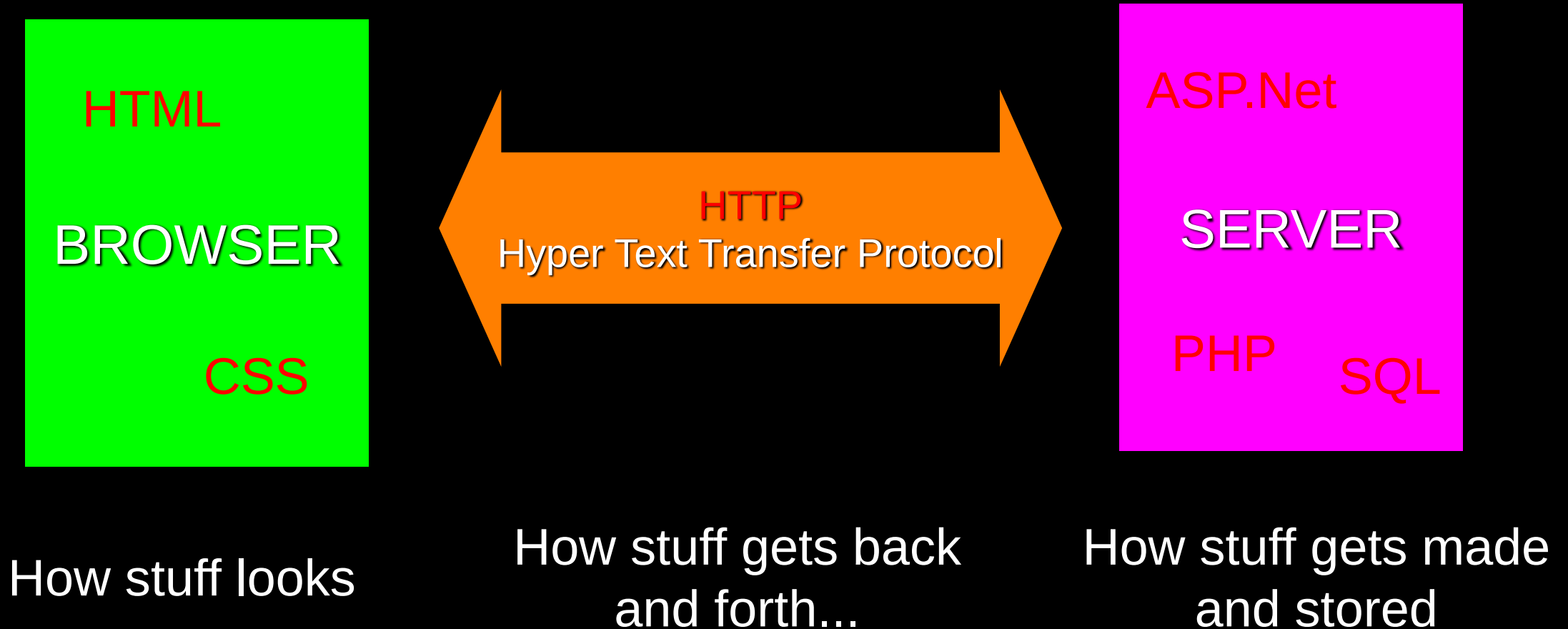


CS 406 – Web Programming

Dr. Atif Aftab Ahmed Jilani
Assistant Professor
atif.jilani@nu.edu.pk

Maintaining State in HTTP



For each of these aspects of the web, we have many **standards and languages** and techniques to learn.

www.umich.edu



www.facebook.com



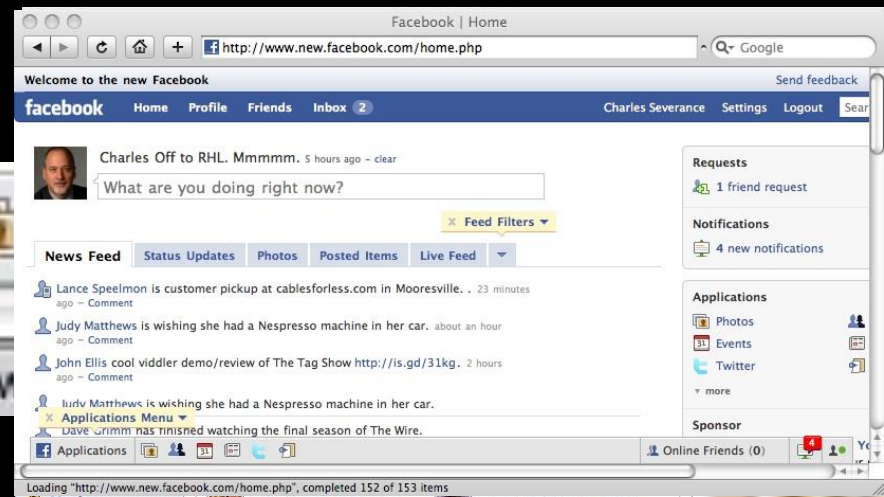
www.yahoo.com



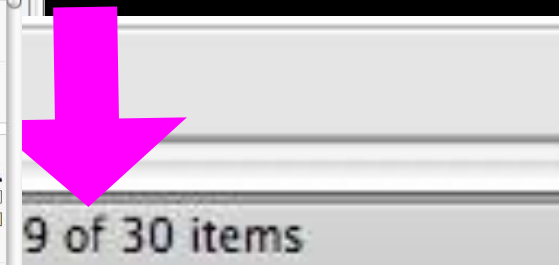
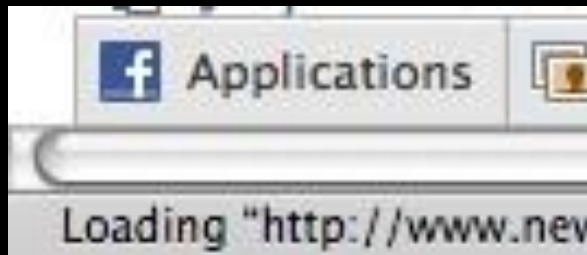
29
times

images.yahoo.com

The read-only web:
hypertext navigation
and lots of **GETs**

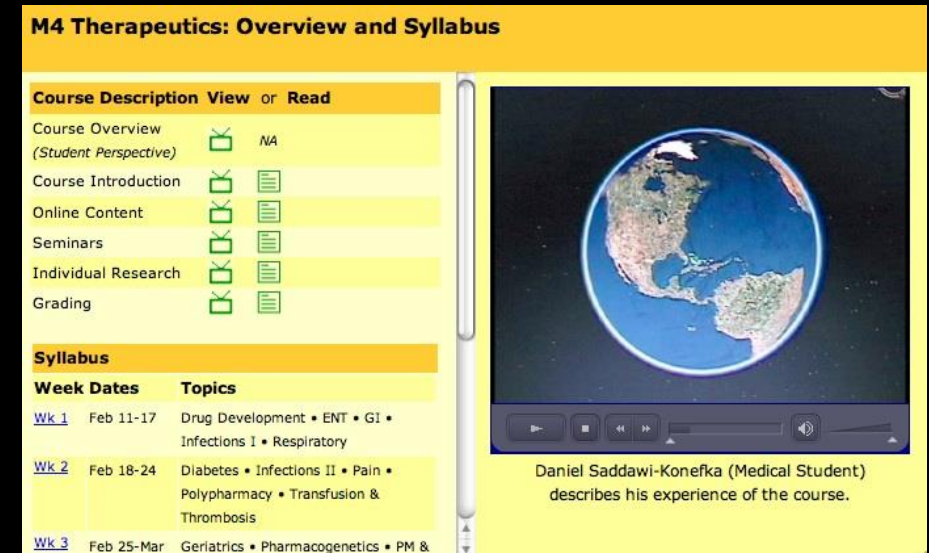
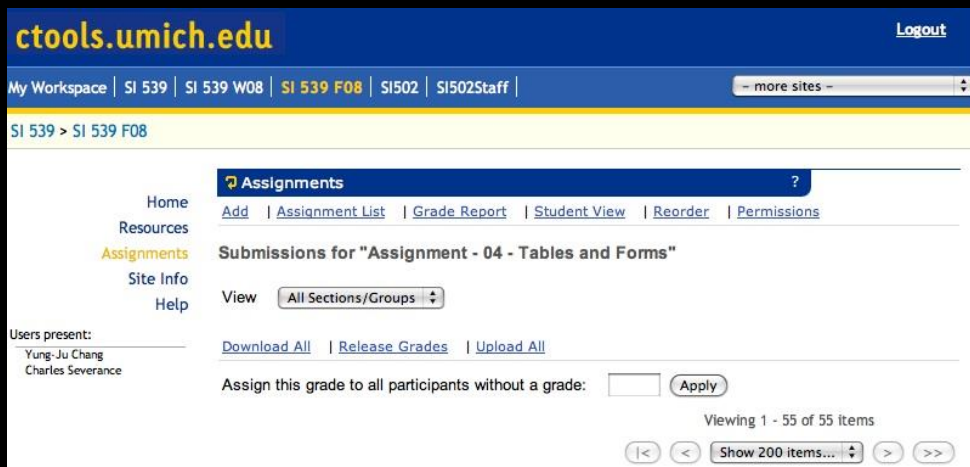
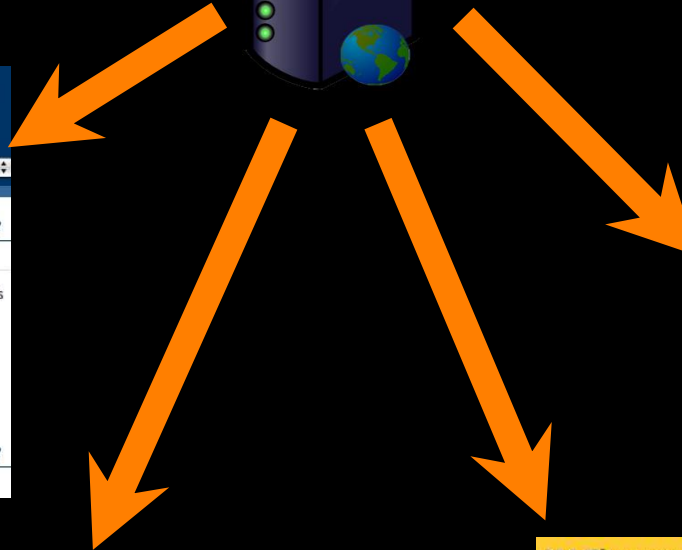


(Screenshot) Source: www.facebook.com
(Globe) source: <http://www.clker.com/clipart-2123.html>
(Server) source: <http://www.clker.com/clipart-server.html>



ctools.umich.edu

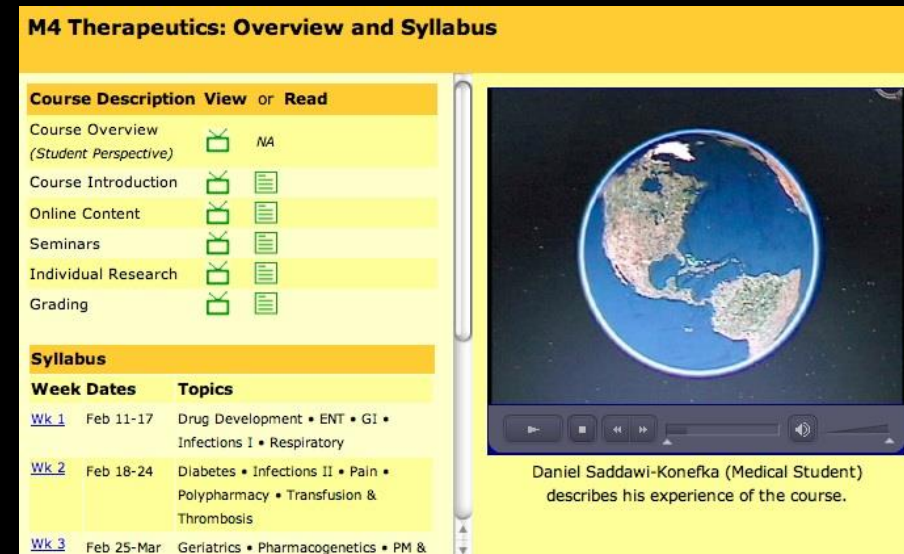
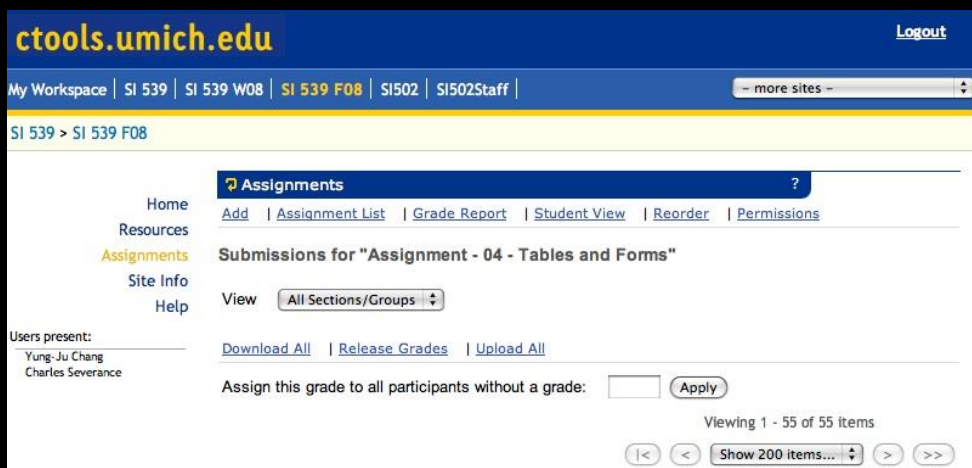
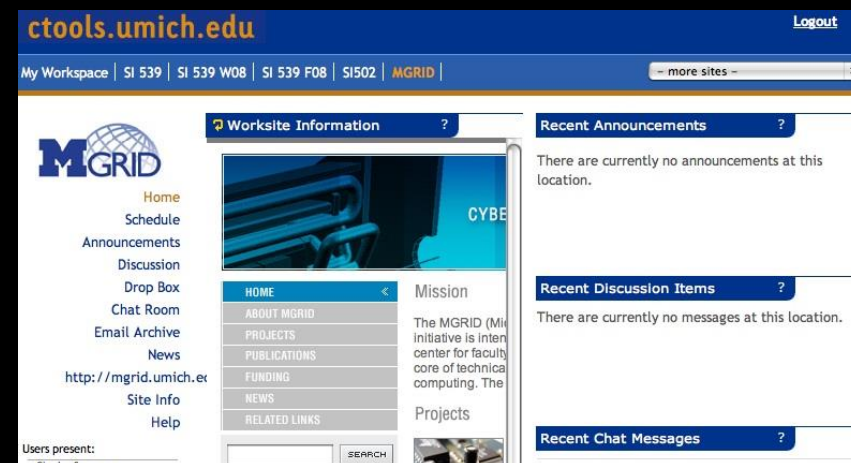
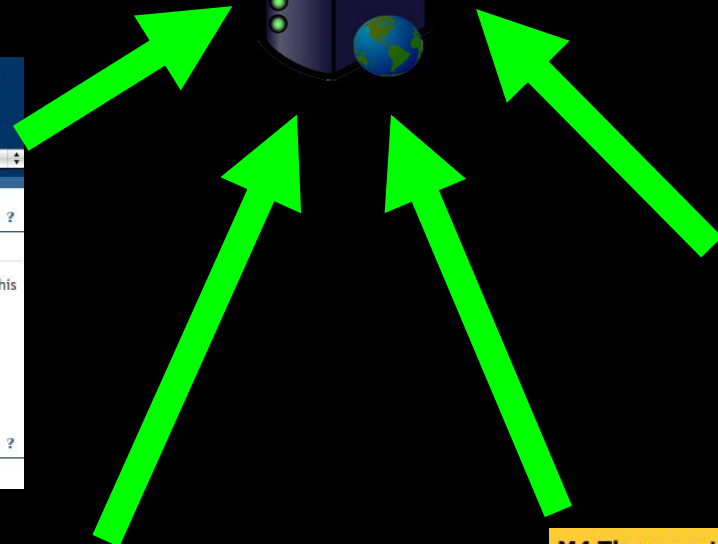
Servers **get** used by many users at the same time.



When folks hit a button...
everyone **POSTs**

???

ctools.umich.edu



??? Server Questions ???

- Who is this user?
- Are they logged in yet?
- What screen did they come from?
- What button did they push?
- Where do we store this data?
- What screen do they want next?

??? Server Questions ???

- 
- Who is this user?
 - Are they logged in yet?
 - What screen did they come from?
 - What button did they push?
 - Where do we store this data?
 - What screen do they want next?

Over and over and
over and over and
over and over and
over....

same as it ever was

Cookies and Sessions

Maintaining State in HTTP

High Level Summary

- The web is “stateless” - the browser does not maintain a connection to the server while you are looking at a page. You may never come back to the same server - or it may be a long time - or it may be one second later
- So we need a way for servers to know “which browser is this?”
 - In the browser state is stored in “Cookies”
 - In the server state is stored in “Sessions”

 UNIVERSITY OF MICHIGAN WEBLOGIN

AUTHENTICATION REQUIRED::

You are connecting to a U-M website that requires authentication. Please enter your Login ID (username or Friend ID) and password to continue.

Need a Login ID?
If you don't have a Login ID, you can [create one now](#).

Login ID

Password

 MToken

Log In

[Forgot your password?](#)

[Login Help](#)

By using this service you agree to adhere to [U-M computing policies and guidelines](#).

Some Web sites always seem to want to know who you are!

What are you doing?

140

Hi, [your profile](#)



drchuck

update

Recent

Replies

Archive



dkeats Spending the morning with the eLearning team innovation, and participation in a community of practice Energizing. 3 minutes ago from web ☆ ↻



drchuck Working on lecture slides – talking about cool sessions. 5 minutes ago from web ☆ 🗑



Matth Ordered Urban Fortunes from Amazon because saying it said things I wasn't at all agreeing with. Right. ago from [twitterrific](#) ☆ ↻



microcline Back from walking the dogs. Cleaned them can, but they don't get to share the futon tonight. 31 m from web ☆ ↻

flickr LOVES YOU TM

[Home](#)

[You](#)

[Organize](#)

[Contacts](#)

[Groups](#)

[Explore](#)

Signed in as dr-chuck

✉ (1 new)

Search everyone's photos



Ni hao dr-chuck!

Now you know how to greet people in Mandarin!

- You have [1 new message](#).
- [Find your friends](#)

Flickr News

31 Mar 08 - Flickr is more fun with friends, but it is a big busy place, and sometimes it can be hard to find the people you know. The new Find Your... [read more news](#)

» [Flickr Blog](#) Great photos & latest news, daily!

ADVERTISEMENT



» [Upload Photos](#) (Or, look at our uploading [tools...](#))

» [Your Photos](#) ([Recent activity](#) / [Comments you've made](#))



» [Photos from your Contacts](#)



From [Lance70](#)



From [Lance70](#)



From [Lance70](#)



From [Lance70](#)

Other Web sites always seem to know who you are!

Browser



Click Draw

Click Draw

You watch the YouTube video
for 30 seconds

Whole
Page

Whole
Page

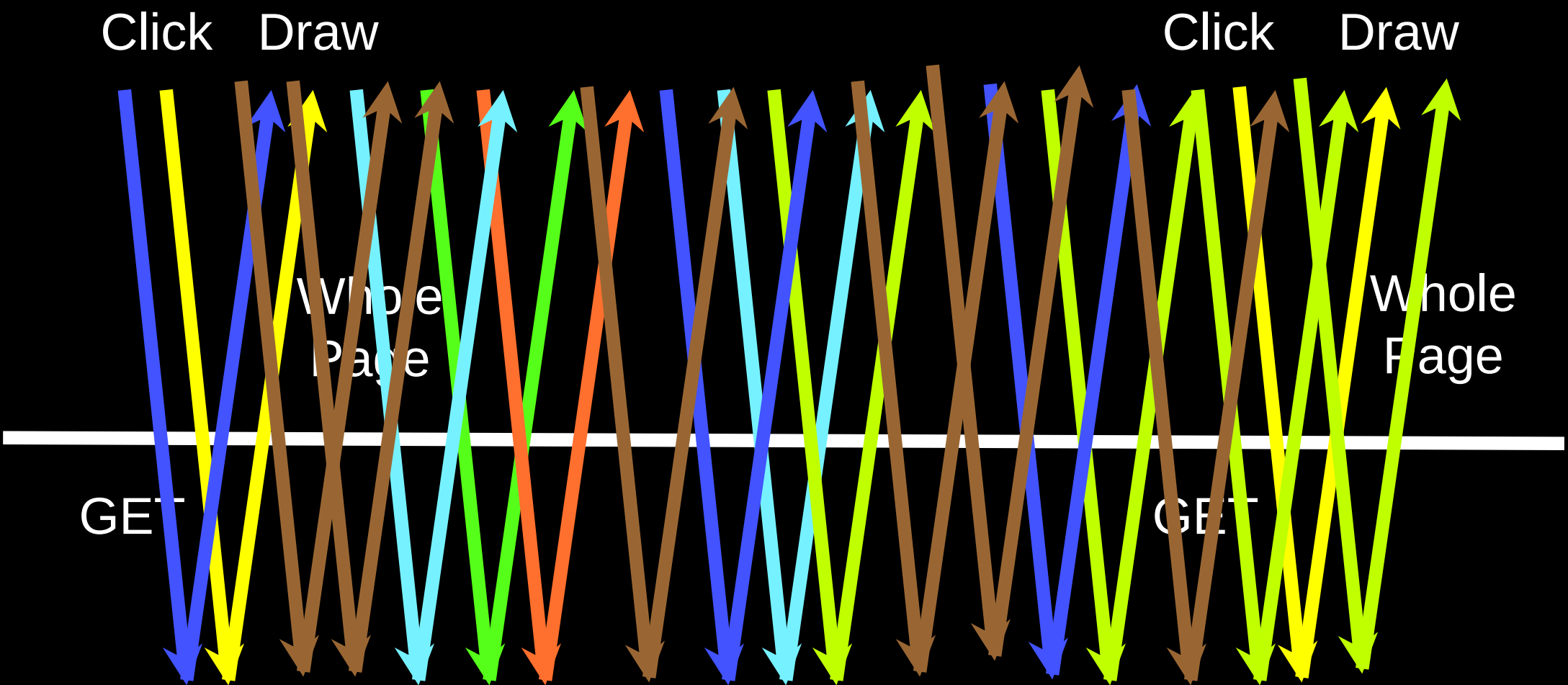
GET

GET

Server

How you see YouTube...

Browser



Server

How YouTube sees you...

Multi-User

- When a server is interacting with many different browsers at the same time, the server needs to know **which** browser a particular request came from
- Request / Response initially was stateless - all browsers looked identical - this was really really bad and did not last very long at all.

Http Get Vs Post

- https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_ajax_get
- https://www.w3schools.com/jquery/tryit.asp?filename=tryjquery_ajax_post

Jquery Get & Post

- https://www.w3schools.com/tags/tryit.asp?filename=tryhtml_form_method
- https://www.w3schools.com/tags/tryit.asp?filename=tryhtml_form_method_post

Web Cookies to the Rescue

Technically, cookies are arbitrary pieces of data chosen by the Web server and sent to the browser. The browser returns them unchanged to the server, introducing a state (memory of previous events) into otherwise stateless HTTP transactions. Without cookies, each retrieval of a Web page or component of a Web page is an isolated event, mostly unrelated to all other views of the pages of the same site.

http://en.wikipedia.org/wiki/HTTP_cookie

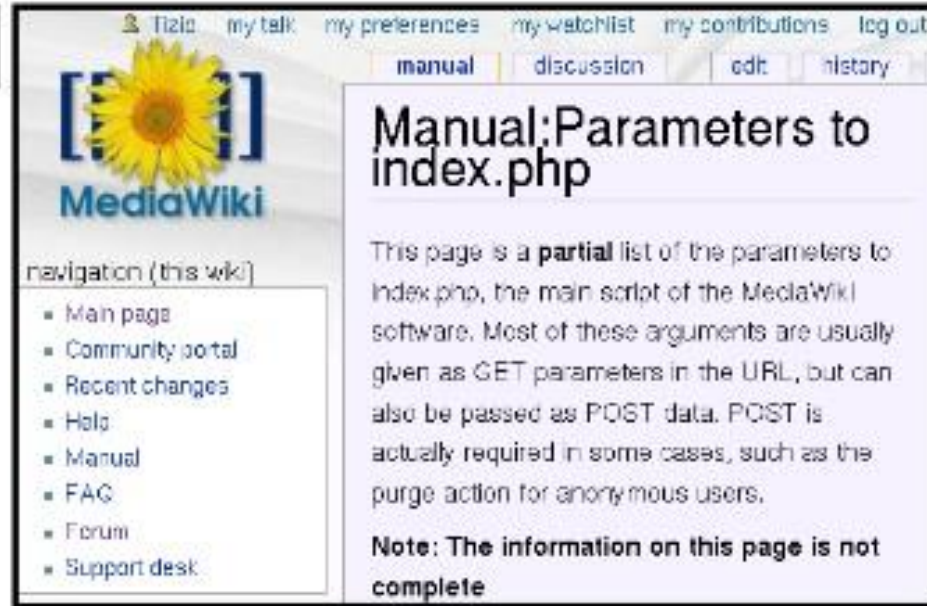
1. browser requests a Web page



2. server sends page+cookie



cookie



server

3. browser requests another page



cookie

Web
browser

http://en.wikipedia.org/wiki/HTTP_cookie

Cookies In the Browser

- Cookies are marked as to the web addresses they come from - the browser only sends back cookies that were originally set by the same web server
- Cookies have an expiration date - some last for years - others are short-term and go away as soon as the browser is closed

Playing with Cookies

- Firefox Developer Plugin has a set of cookie features
- Other browsers have a way to view or change cookies

Home

Worksite Setup

Membership

Schedule

Announcements

Resources

News

Web Content

Preferences

Teaching
Questionnaires

Help

Users present:

Charles Severance

Message of the Day

[Options](#)

Instructors: Textbook Access

Now available, class instructors can enter by October 29 early order de

Teaching Questionnaire Sullivan

I am writing to tell you about an in most of the University's schools and this semester, Fall 2008, TQ evaluation longer be printed. [Instructors read](#)

Move, Hide, and Rearrange

You can customize your tabs using 'Customize Tabs' item in the Pref

Cookie Information - https://ctools.umich.edu/portal

Sakai dr-chuck.com NWA Chuck's Media iPhone Navigation I3k CT SI 182 Sakai Based Service : ... portal

Disable Cookies CSS Forms Images Information Miscellaneous Outline Resize To

CTools : My Workspace : Home Cookie Information - https://ct...

NAME	__utmz
VALUE	54743033.1224729949.7.7.utmcsr=rubylearn.com utmccn=(referral) utmcmd=referrallutmct=/
HOST	.ctools.umich.edu
PATH	/
SECURE	No
EXPIRES	Thu, 23 Apr 2009 14:45:49 GMT

Edit Cookie
Delete Cookie

NAME	JSESSIONID
VALUE	2962baf3-60fb-42cb-a05d-6213574eb7c8.kungpao
HOST	ctools.umich.edu
PATH	/
SECURE	No
EXPIRES	At End Of Session

Done

Two Kinds of Cookies

- Two kinds of cookies
 - Long-lived - who you are - account name last access time - you can close and reopen your browser and it is still there
 - Temporary - used to identify your session - it goes away when you close the browser

Cookie Information - http://www.dr-chuck.com/

Sakai dr-chuck.com NWA Chuck's Media iPhone Navigation I3k CT SI 182 Sakai Based Service : ... portal

Disable Cookies CSS Forms Images Information Miscellaneous Outline Resize Tools

Dr. Chuck's Interactive Person... Cookie Information - http://ww...

NAME	__utmb
VALUE	140904167
HOST	.dr-chuck.com
PATH	/
SECURE	No
EXPIRES	Wed, 24 Sep 2008 10:28:02 GMT

[Edit Cookie](#)

[Delete Cookie](#)

NAME	__utmc
VALUE	140904167
HOST	.dr-chuck.com
PATH	/
SECURE	No
EXPIRES	At End Of Session

Done

The Firefox Web Developer Plugin Shows Cookies for the Current Host.

Cookie Information - http://www.dr-chuck.com/

Sakai dr-chuck.com NWA Chuck's Media iPhone Navigation I3k CT SI 182 Sakai Based Service : ... portal

Disable Cookies CSS Forms Images Information Miscellaneous Outline Resize Tools

Dr. Chuck's Interactive Person... Cookie Information - http://ww...

NAME	__utmb
VALUE	140904167
HOST	.dr-chuck.com
PATH	/
SECURE	No
EXPIRES	Wed, 24 Sep 2008 10:28:02 GMT

[Edit Cookie](#)

[Delete Cookie](#)

Google Analytics Cookies

NAME	__utmc
VALUE	140904167
HOST	.dr-chuck.com
PATH	/
SECURE	No
EXPIRES	At End Of Session

Done

Request Response Again!
This time with cookies...

HTTP Request / Response Cycle

(Review)

Web Server

HTTP
Request

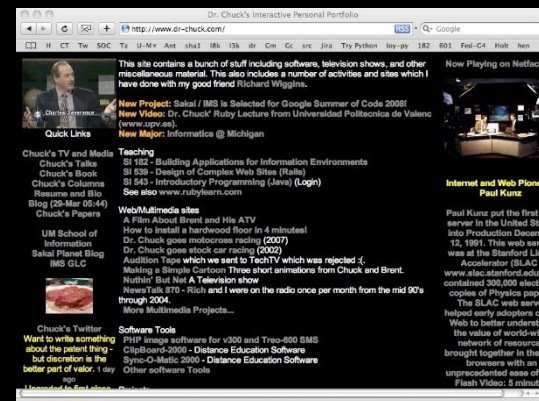
HTTP
Response

Browser

Internet Explorer,
FireFox, Safari, etc.

Hello there my name is Chuck
Go ahead and click on [here](#).

http://www.oreilly.com/openbook/cgi/ch04_02.html

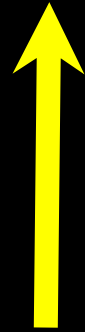


HTTP Request / Response Cycle

Web Server

GET /index.html HTTP/1.1
Accept: www/source
Accept: text/html
User-Agent: Lynx/2.4

HTTP
Request



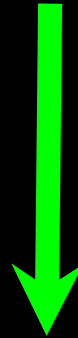
Browser

We do our initial GET to a server. The server checks to see if we have a cookie with a particular name set. Since this is our first interaction, we do not have cookies set for this host.

HTTP Request / Response Cycle

Along with the rest of the response, the server sets a cookie with some name (**sessid**) and sends it back along with the rest of the response.

Web Server



Browser

HTTP/1.1 200 OK
Content-type: text/html
Set-Cookie: **sessid=123**

<head> .. </head>
<body>
<h1>Welcome

HTTP
Response

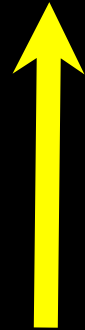
host: sessid=123

HTTP Request / Response Cycle

Web Server

GET /index.html HTTP/1.1
Accept: www/source
Accept: text/html
Cookie: sessid=123
User-Agent: Lynx/2.4

HTTP
Request

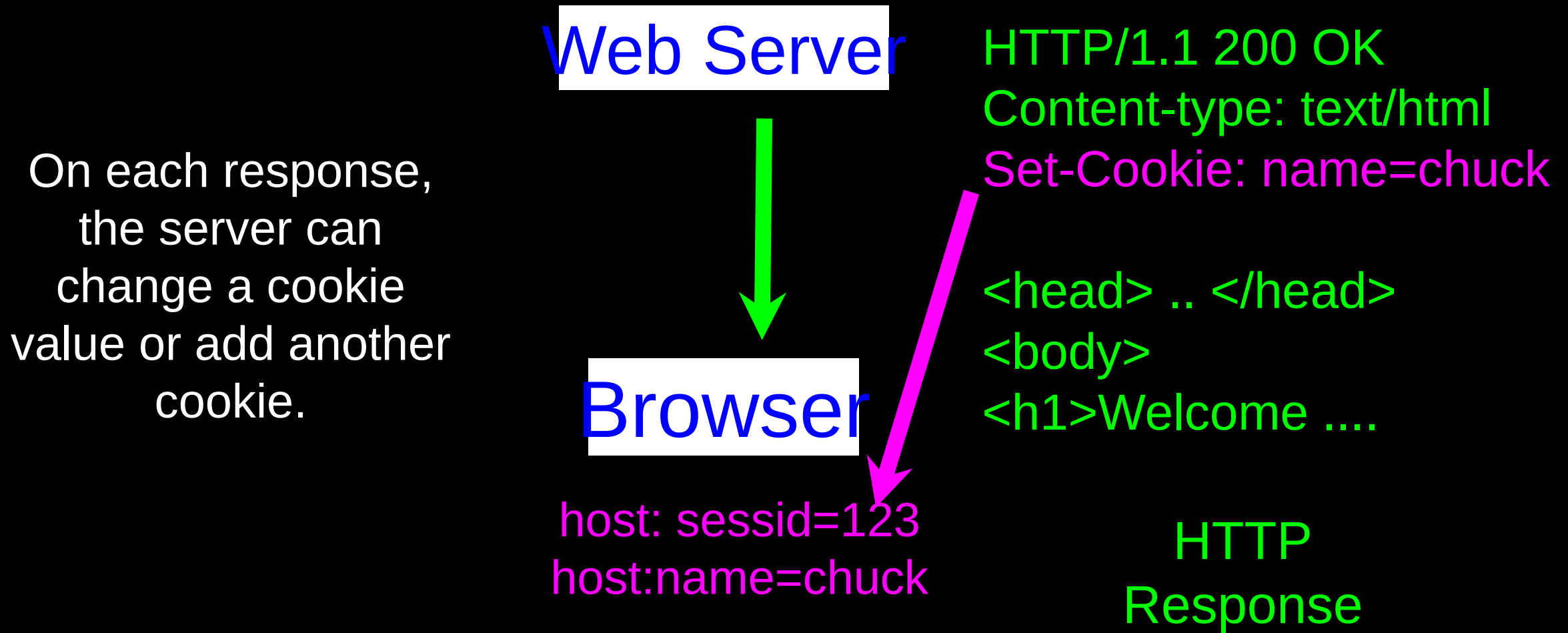


Browser

host: sessid=123

From that point forward, each time we send a GET or POST to the server, we include any cookies which were set by that host.

HTTP Request / Response Cycle



HTTP Request / Response Cycle

Web Server

GET /index.html HTTP/1.1
Accept: www/source
Accept: text/html
Cookie: sessid=123,name=chuck
User-Agent: Lynx/2.4

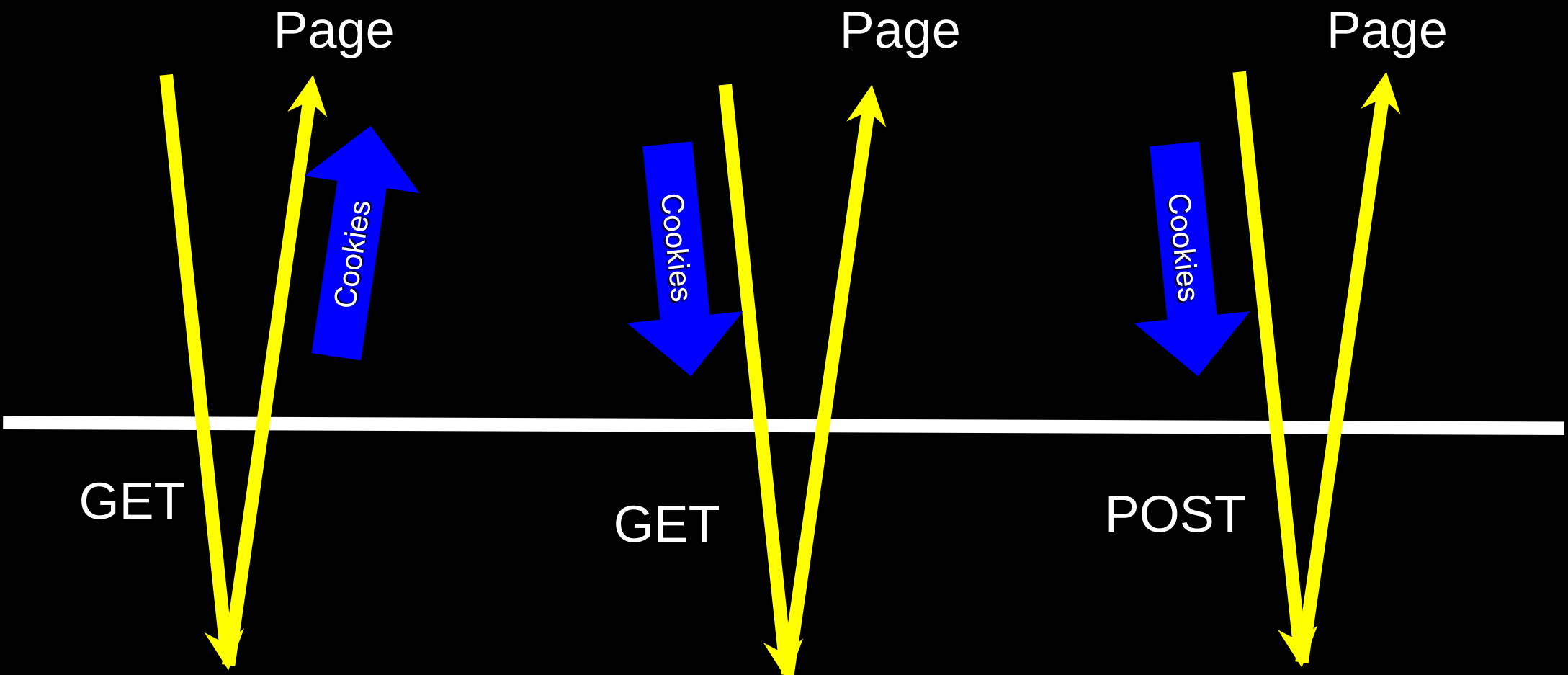
HTTP
Request

Browser

host: sessid=123
host: name=chuck

From that point forward, each time we send a GET or POST to the server, we include all the cookies which were set by that **host**.

Browser



Server

Remember that cookies are only sent back to the host that set the cookie.

Security

- We only send cookies back to the host that originally set the cookie
- The browser has *lots* of cookies for lots of hosts
- To see all Cookies: Firefox -> Preferences -> Privacy -> Show Cookies



Using Cookies to Support Sessions and Login / Logout

 UNIVERSITY OF MICHIGAN WEBLOGIN

AUTHENTICATION REQUIRED::

You are connecting to a U-M website that requires authentication. Please enter your Login ID (username or Friend ID) and password to continue.

Need a Login ID?
If you don't have a Login ID, you can [create one now](#).

Login ID

Password

▶ MToken

Log In

[Forgot your password?](#)
[Login Help](#)

By using this service you agree to adhere to [U-M computing policies and guidelines](#).

Some Web sites always seem to want to know who you are!

In The Server - Sessions

- In most server applications, as soon as we meet a new browser - we create a session
- We set a session cookie to be stored in the browser which indicates the session id in use
- The creation and destruction of sessions is generally handled by a web framework or some utility code that we just use to manage the sessions

Session Identifier

- A large, random number that we place in a browser cookie the first time we encounter a browser.
- This number is used to pick from the many sessions that the server has active at any one time.
- Server software stores data in the session which it wants to have from one request to another from the same browser.
- Shopping cart or login information is stored in the session in the server

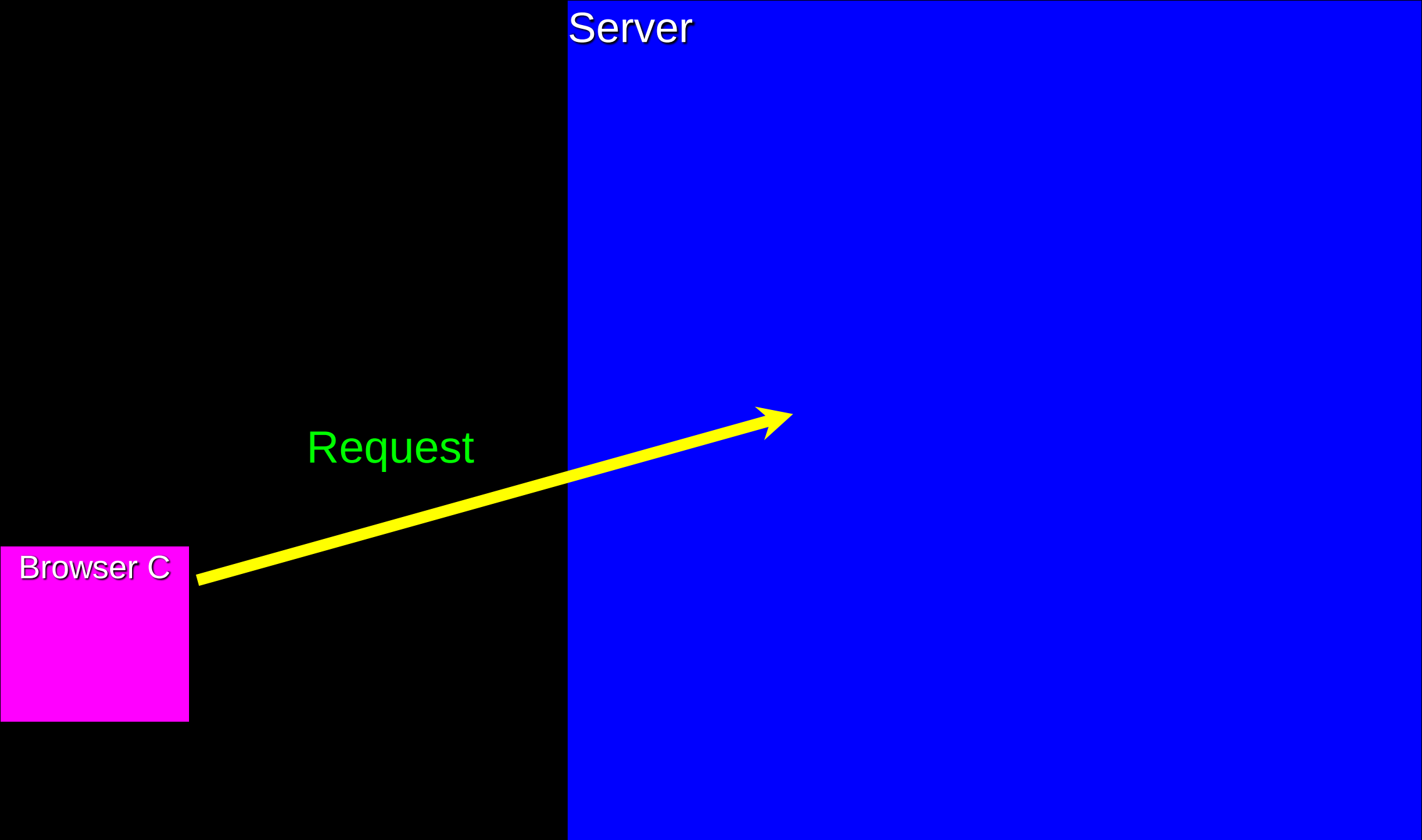
Server

Browser C

Server

Request

Browser C



```
graph LR; C[Browser C] -- Request --> S[Server]
```

The diagram illustrates a network communication. On the left, a small red square represents 'Browser C'. On the right, a large blue rectangle represents the 'Server'. A thick blue arrow points from the right side of the red square to the left side of the blue rectangle. The word 'Request' is written in red text above the arrow, indicating the nature of the communication.

Server

Session 97

Create
Session

Request

Browser C

cook=97

cook=97

Response

index:

"Please
log in"

Typing



Browser C

cook=97

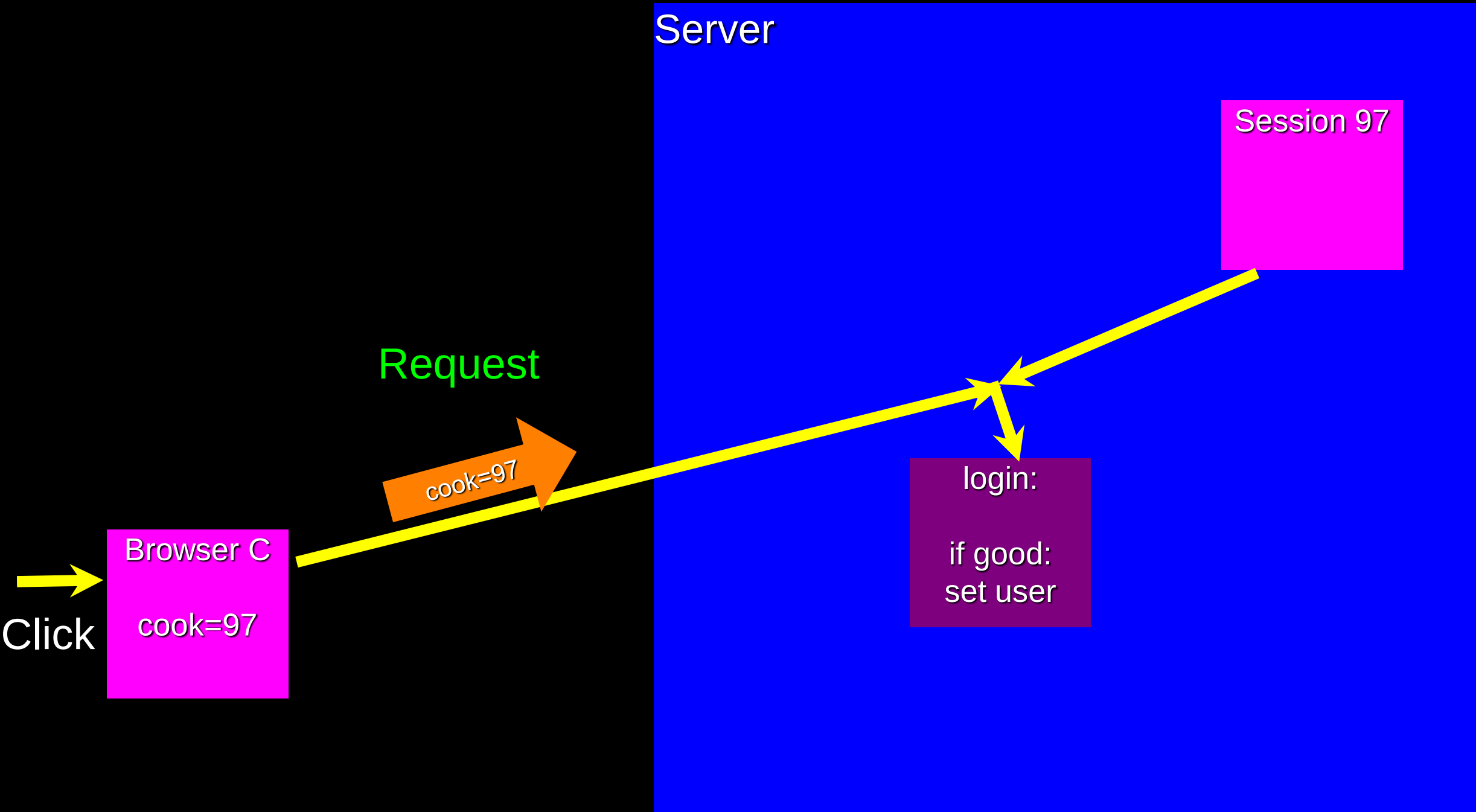
Server

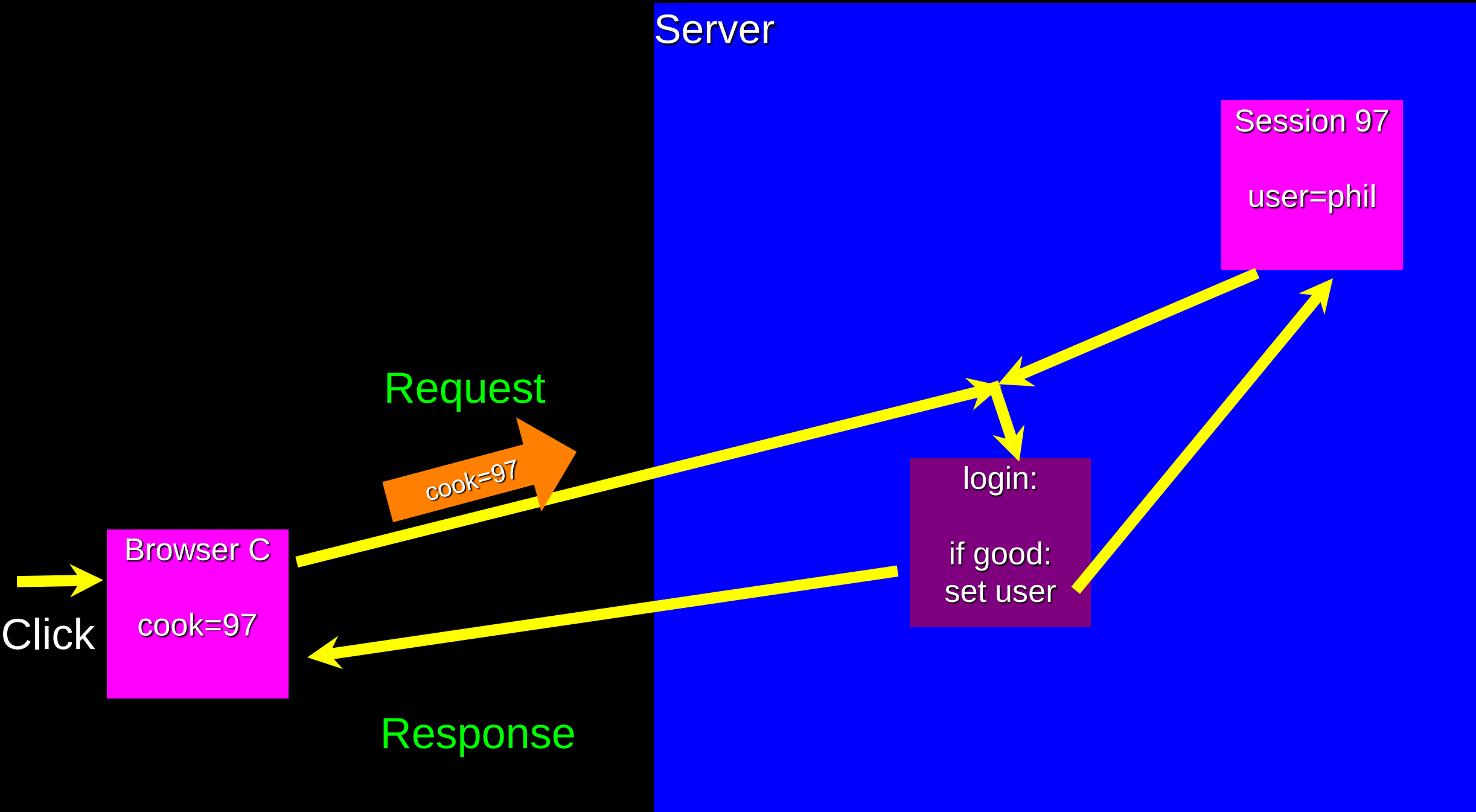
Session 97

We now have a session established but are not yet logged in.

Login / Logout

- Having a session is not the same as being logged in.
- Generally you have a session the instant you connect to a web site
- The Session ID cookie is set when the first page is delivered
- Login puts user information in the session (stored in the server)
- Logout removes user information from the session





Server

Session 97

user=phil

Browser C

cook=97

Using Sessions for Other Stuff

Browser A

cook=10

Browser B

cook=46

Server

Session 10

user=chuck
bal=\$1000

Session 46

user=jan
bal=\$400

Browser A

cook=10

Browser B

cook=46

Server

Session 10

user=chuck
bal=\$1000

Session 46

user=jan
bal=\$500

withdraw:

bal=bal-100

Browser A

cook=10

Browser B

cook=46

Click

Server

Session 10

user=chuck
bal=\$1000

Session 46

user=jan
bal=\$500

withdraw:

bal=bal-100

Server

Browser A

cook=10

Browser B

cook=46

Session 10

user=chuck
bal=\$1000

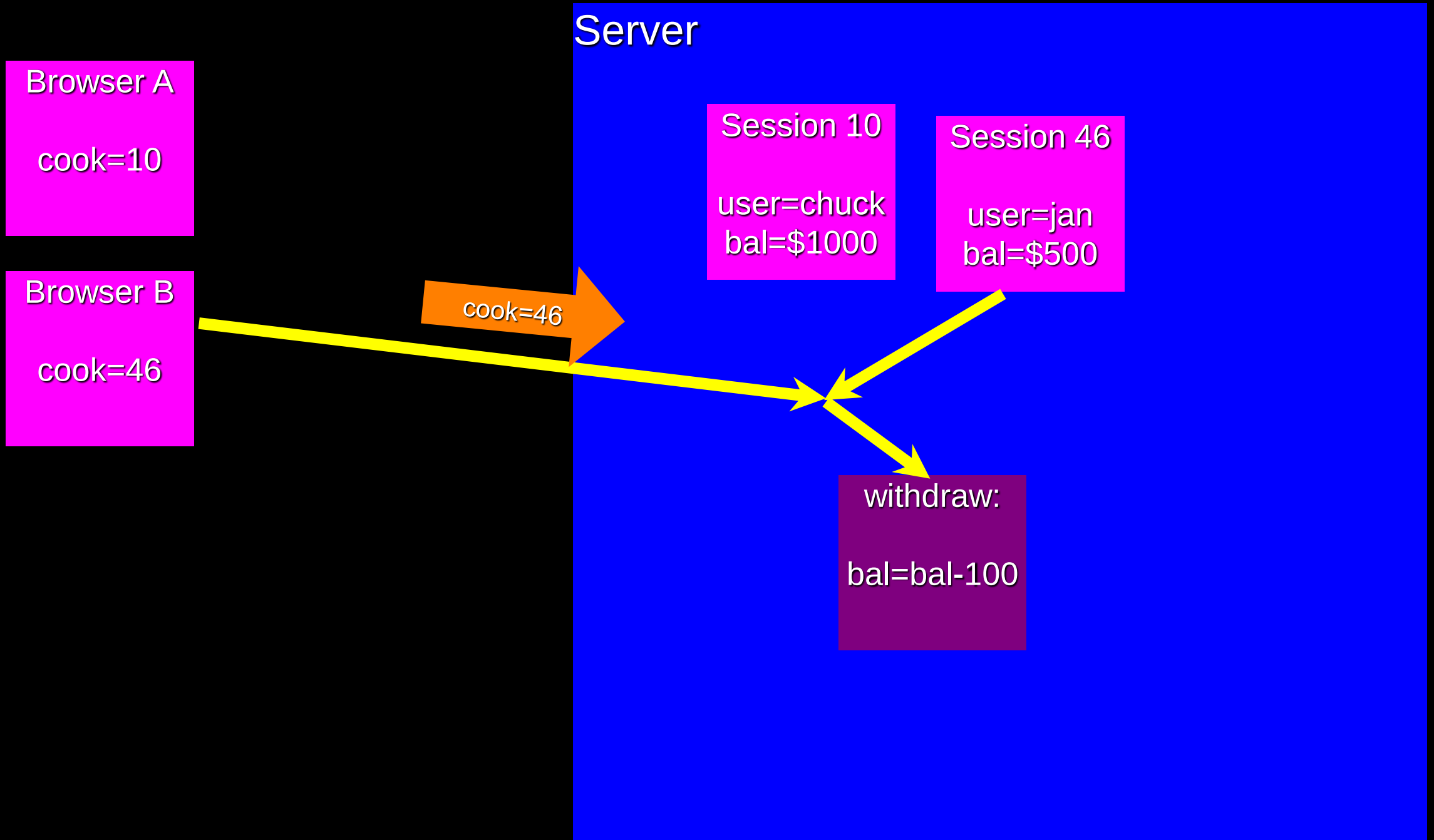
Session 46

user=jan
bal=\$500

cook=46

withdraw:

bal=bal-100



Server

Browser A
cook=10

Browser B
cook=46

Request

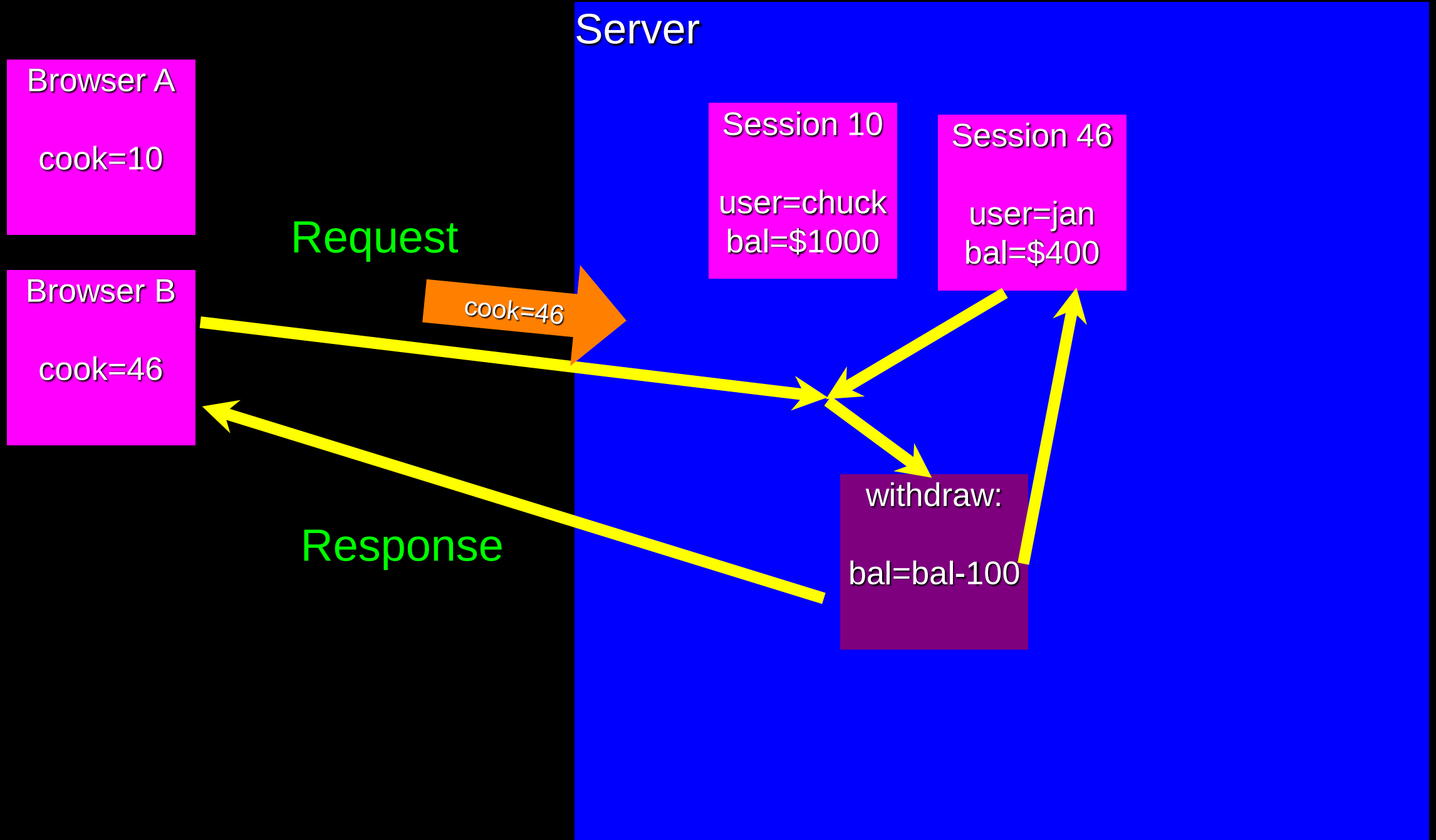
cook=46

Session 10
user=chuck
bal=\$1000

Session 46
user=jan
bal=\$400

Response

withdraw:
bal=bal-100



Review...

High Level Summary

- The web is “stateless” - the browser does not maintain a connection to the server while you are looking at a page. You may never come back to the same server - or it may be a long time - or it may be one second later
- So we need a way for servers to know “which browser is this?”
 - In the browser state is stored in “Cookies”
 - In the server state is stored in “Sessions”

Browser



Click Draw

Click Draw

You watch the YouTube video
for an 30 seconds

Whole
Page

Whole
Page

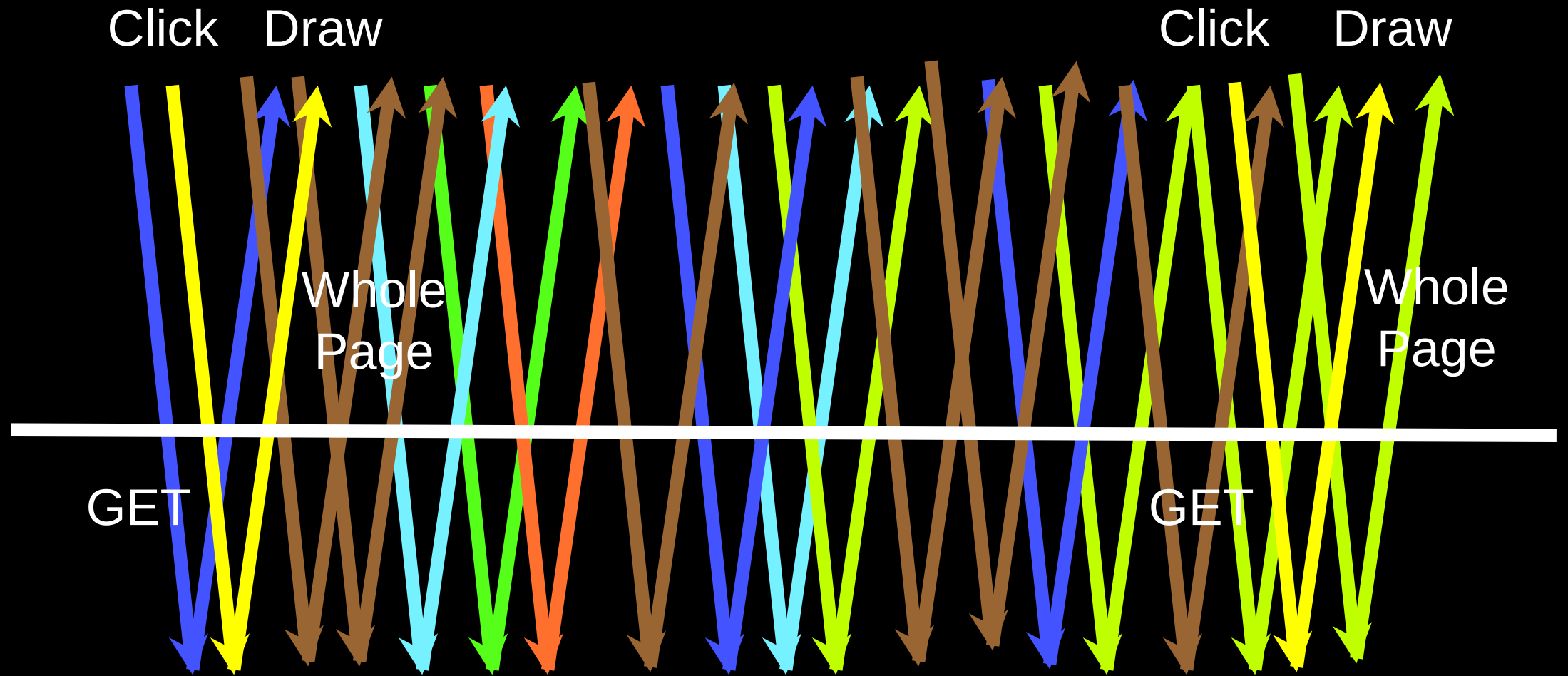
GET

GET

Server

How you see YouTube...

Browser



Server

Browser

Click Draw

Click Draw

Whole
Page

Whole
Page

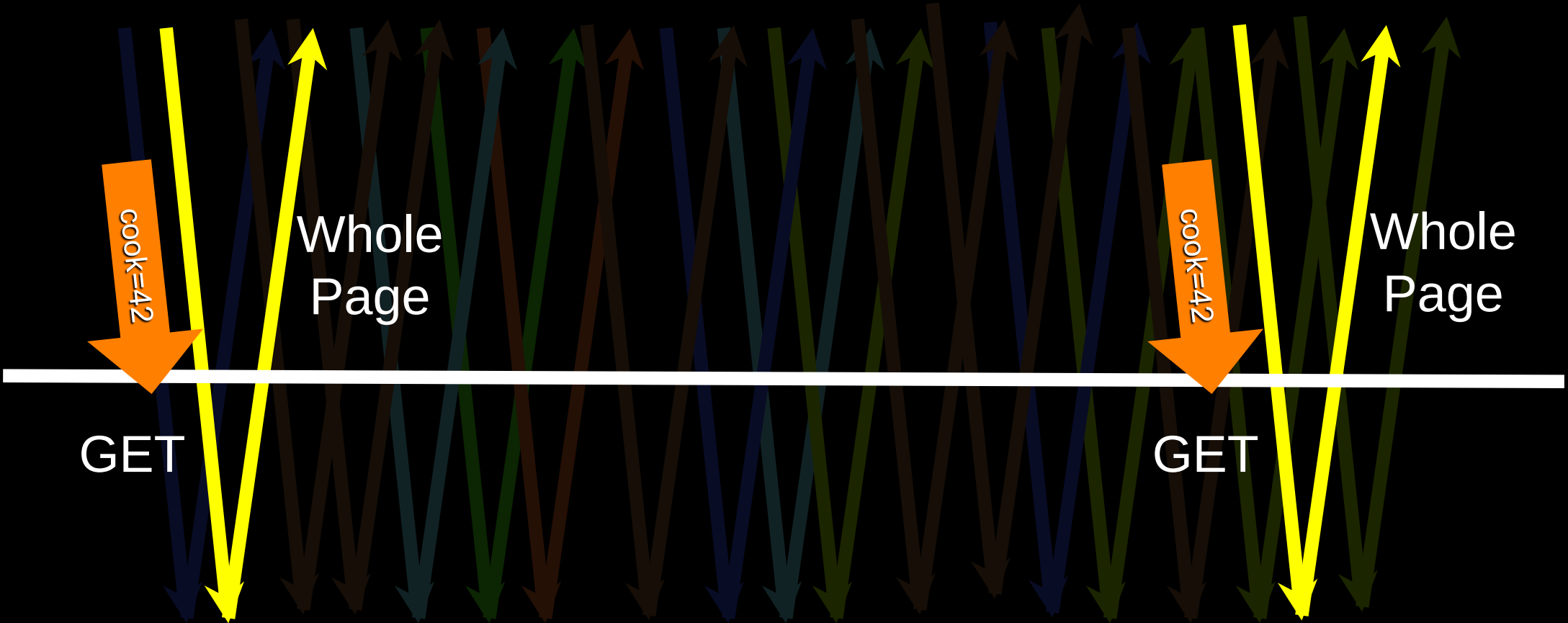
GET

GET

Session 42

Session 42

Server



??? Server Questions ???

- Who is this user?
- Are they logged in yet?
- What screen did they come from?
- What button did they push?
- Where do we store this data?
- What screen do they want next?

Cookie/Session Summary

- Cookies take the stateless web and allow servers to store small “breadcrumbs” in each browser.
- Session IDs are large random numbers stored in a cookie and used to maintain a session on the server for each of the browsers connecting to the server
- Server software stores sessions *somewhere* - each time a request comes back in, the right session is retrieved based on the cookie
- Server uses the session as a scratch space for little things

Acknowledgement

- Charles Severance
- www.dr-chuck.com

Thank You